

Day 2: Relativistic Kinematics I & II: Rapidity & Pseudorapidity in Detail

14-Day Hands-on Course on Heavy-Ion Phenomenology

Lecture

NIT-J, HEP-119

June 2026

Syllabus: 14-Day Hands-on Course Layout

Part I: Relativistic Kinematics

- Day 1: Foundations of Relativistic Kinematics & Experimental HEP
- **Day 2: Kinematics I & II: Rapidity & Pseudorapidity in Detail**
- Day 3: Kinematics III: Invariant Yield & Phase Space Jacobian
- Day 4: Kinematics IV: Coordinate Calculator implementations

Part II: AMPT Physics & Mechanics

- Day 5: AMPT Evolution: HIJING, ZPC, Hadronization, ART
- Day 6: AMPT Data format: event headers & parsing
- Day 7: Single-Particle Observables: Multiplicity & p_T

Part III: Thermodynamics & Collectivity

- Day 8: Transverse Spectra, Identified Particles & m_T Scaling
- Day 9: Centrality & Geometry: Glauber vs AMPT
- Day 10: Collective Flow: Elliptic Flow (v_2)
- Day 11: Medium Modification: ZPC Knobs & Mean Free Path

Part IV: Advanced Scans & Pipeline

- Day 12: Energy Scan: Beam Energy Scan (STAR at RHIC)
- Day 13: Advanced Analysis
- Day 14: Wrap-Up: Reproducible HEP Data Pipeline

Polar Angle to Pseudorapidity Mapping

Theoretical Definition (Sahoo Eq. 5.138)

At ultra-relativistic limits ($p \gg m_0$), the momentum rapidity y simplifies to a purely angular, geometric quantity called **Pseudorapidity** (η):

$$\eta = -\ln \tan \left(\frac{\theta}{2} \right)$$

Where θ is the polar angle relative to the beam axis (z).

Physical Advantages of Pseudorapidity

- **No Particle ID (PID) needed:** Calculated solely from the emission angle θ of the trajectory in the tracker.
- **No magnetic field needed:** Does not require momentum information (only spatial tracking lines).
- **Symmetric / Odd behavior:** It is odd about $\theta = 90^\circ$ ($\eta = 0$). That is, $\eta(180^\circ - \theta) = -\eta(\theta)$, reflecting forward-backward symmetry.

Angular Detector Mapping: θ vs. η

Angle to η Reference (Sahoo Table 5.3)

Polar Angle θ	Pseudorapidity η	Detector Region
0°	∞	Forward Beam pipe
10°	2.44	Endcap region
30°	1.32	Forward transition
45°	0.88	Diagonal transition
90°	0.00	Central Barrel
135°	-0.88	Diagonal transition
170°	-2.44	Backward transition
180°	$-\infty$	Backward Beam pipe

Physical Trajectories (Sahoo Fig. 5.13)

- $\theta = 90^\circ$ ($\eta = 0$): Particles fly straight out transversely.
- $\theta \rightarrow 0^\circ$ ($\eta \rightarrow \infty$): Particles fly forward along the beam pipe.
- High $|\eta|$ values denote regions extremely close to the beam line (demanding high spatial resolution trackers).

Interactive Physics Simulator

Open in your browser:

- animations/05_pseudo-rapidity_detector_angle.html

(Interactive θ vs. η detector mapping)

Coordinate Transformation: From (y, p_T) to (η, p_T)

Vector Decompositions (Sahoo Eq. 5.141 – 5.143)

By defining the pseudorapidity η in terms of longitudinal momentum p_z and total momentum magnitude $|\mathbf{p}|$:

$$e^{\eta} = \sqrt{\frac{|\mathbf{p}| + p_z}{|\mathbf{p}| - p_z}}, \quad e^{-\eta} = \sqrt{\frac{|\mathbf{p}| - p_z}{|\mathbf{p}| + p_z}}$$

We can solve for the momentum component variables in terms of (p_T, η) :

$$|\mathbf{p}| = p_T \cosh \eta, \quad p_z = p_T \sinh \eta$$

Which allows us to write the energy E and transverse mass m_T elegantly:

$$E = \sqrt{|\mathbf{p}|^2 + m_0^2} = \sqrt{p_T^2 \cosh^2 \eta + m_0^2}$$

The Phase-Space Jacobian: Mathematical Definition

Rapidity to Pseudorapidity Mapping (Sahoo Eq. 5.147)

The distribution of particles as a function of pseudorapidity ($dN/d\eta$) is related to the rapidity distribution (dN/dy) via the phase-space Jacobian $J(y \rightarrow \eta) = dy/d\eta$:

$$\frac{dN}{d\eta dp_T} = J(y \rightarrow \eta) \frac{dN}{dy dp_T} = \sqrt{1 - \frac{m_0^2}{m_T^2 \cosh^2 y}} \frac{dN}{dy dp_T}$$

In the ultra-relativistic (massless) limit $m_0 \rightarrow 0$, the Jacobian simplifies to the particle's longitudinal velocity: $J(y \rightarrow \eta) \rightarrow p/E = \beta \rightarrow 1$.

Integration Over Transverse Momentum

To obtain the overall multiplicity distribution $dN/d\eta$, we must integrate over the transverse momentum p_T :

$$\frac{dN}{d\eta} = \int_0^\infty J(y \rightarrow \eta, p_T) \frac{dN}{dy dp_T} dp_T$$

Because J depends on p_T (through $m_T = \sqrt{p_T^2 + m_0^2}$), the mapping varies strongly with the transverse momentum of the tracks.

The Phase-Space Jacobian: Physical Consequences

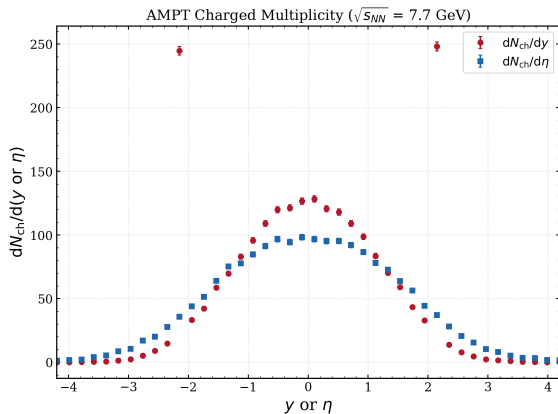
Physical Limits of the Jacobian Factor

- **Massless limit** ($m_0 \rightarrow 0$ or $p \gg m_0$): $J(y \rightarrow \eta) \rightarrow 1 \implies dN/d\eta \approx dN/dy$. This is highly accurate for photons, electrons, or very high p_T hadrons (where mass is negligible).
- **Massive limit** ($m_0 > 0$): $J(y \rightarrow \eta) < 1$ at mid-rapidity ($y \approx 0$). This creates a local, kinematic suppression in the pseudorapidity distribution $dN/d\eta$ even when the underlying physical rapidity distribution dN/dy is flat or peak-like.

Takeaway

The difference between dN/dy and $dN/d\eta$ is a purely mathematical consequence of the change of variables. Detector trackers measure η directly, but physics analyses must correct back to y using the Jacobian factor to access the true dynamics.

AMPT: dN/dy vs. $dN/d\eta$ at $\sqrt{s_{NN}} = 7.7$ GeV



Symmetric Collision at $\sqrt{s_{NN}} = 7.7$ GeV

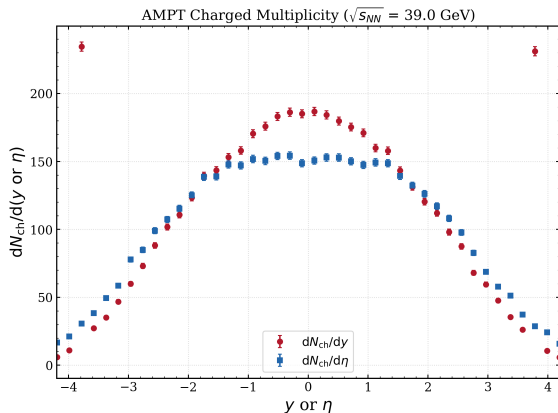
- Represents the lower energy regime of the RHIC Beam Energy Scan.
- The rapidity range is restricted ($y_{\text{beam}} \approx 2.09$), resulting in a narrow distribution.
- Peak charged particle yield per event is relatively small ($dN/dy \approx 45$).
- **Jacobian Suppression:** The suppression at $\eta = 0$ is visible as a flattening of the pseudorapidity peak compared to the rapidity distribution.

Interactive Jacobian Explorer

Open in your browser:

• day2/day2_rapidity_jacobian_explorer.html

AMPT: dN/dy vs. $dN/d\eta$ at $\sqrt{s_{NN}} = 39$ GeV



High Energy Collision at $\sqrt{s_{NN}} = 39$ GeV

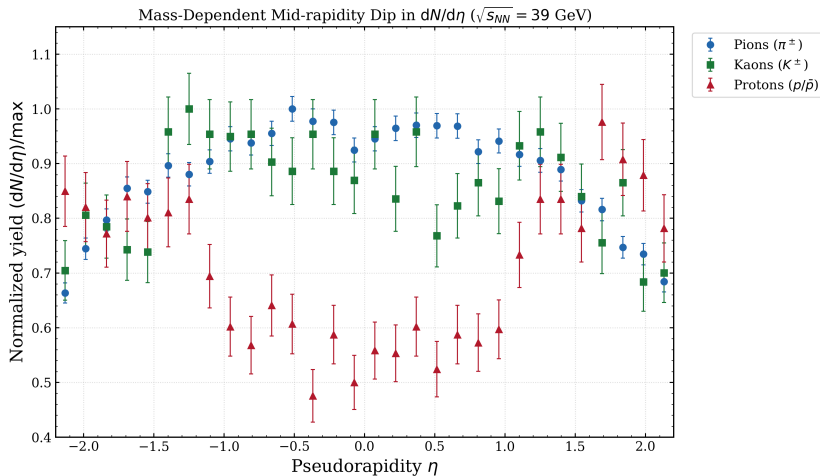
- High energy regime with a wider longitudinal phase space ($y_{\text{beam}} \approx 3.73$).
- Peak yield is significantly higher ($dN/dy \approx 120$) due to increased particle production.
- The difference between dN/dy and $dN/d\eta$ is more pronounced across a broader range, showing a wide plateau in pseudorapidity.

Interactive Jacobian Explorer

Open in your browser:

• day2/day2_rapidity_jacobian_explorer.html

Mass-Dependent Mid-rapidity Dip in $dN/d\eta$



Mass-Dependent Mid-rapidity Dip: The Jacobian Mechanism

Why do heavy particles dip at $\eta = 0$?

At mid-rapidity ($\eta = 0 \iff y = 0$), the Jacobian simplifies to: $J(0) = \frac{p_T}{m_T} = \frac{p_T}{\sqrt{p_T^2 + m_0^2}}$.

- **Pions** ($\pi^\pm$, $m_0 \approx 0.14$ GeV): Very light mass. The ratio $p_T/m_T \approx 1$ for almost all p_T . Thus, $J(0) \approx 1$ (no dip).
- **Kaons** ($K^\pm$, $m_0 \approx 0.49$ GeV): Medium mass. Suppression at low p_T is significant, creating a shallow dip.
- **Protons** ($p/\bar{p}$, $m_0 \approx 0.94$ GeV): Heavy mass. Suppression is severe ($J(0) \ll 1$ for $p_T < m_0$), creating a deep, pronounced mid-rapidity dip.

Physical Takeaway

This dip is purely a coordinate mapping artifact, not physical dynamics (since dN/dy remains flat or peak-like at $y = 0$). Heavy particles have a large rest mass, making their velocity $\beta \ll 1$ at low p_T , which suppresses $dN/d\eta$ via $J \rightarrow \beta$.

Rapidity Boosts: Lab Frame vs. Center-of-Mass Frame

Rapidity Additivity (Sahoo Eq. 5.43)

Under a collinear Lorentz boost of velocity β_b (boost rapidity y_b), the rapidity transforms linearly: $y' = y - y_b$. This shifts the entire distribution dN/dy' as a rigid body, preserving its shape and width σ_y .

Relativistic Beaming (Searchlight Effect)

At high boosts, particles are beamed forward (in the direction of the boost). This increases the local density at high η' , showing polar angle bunching.

Pseudorapidity Deformation

Pseudorapidity η does **not** transform additively under boosts because the polar angle θ transforms non-linearly. The distribution $dN/d\eta'$ skews and deforms.

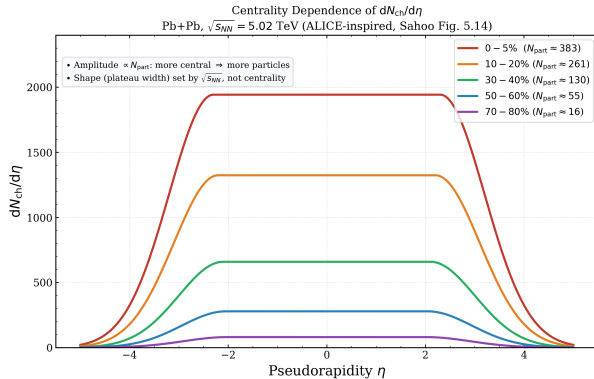
Interactive Boost Simulator

Open in your browser:

- day2/day2_lab_vs_cms_rapidity.html

(Observe rapidity shift vs pseudorapidity deformation and beaming)

Centrality Dependence of Pseudorapidity Distributions



Collision Geometry & Glauber Model

Multiplicity **amplitude** scales with the overlapping nuclear volume:

- **Impact Parameter (b)**: Distance between nuclear centers. Central ($b \approx 0$ fm, 0 – 5%) vs. Peripheral ($b > 12$ fm, > 70%).
- **Participants (N_{part})**: Nucleons undergoing ≥ 1 inelastic collision. Directly sets the fireball volume and multiplicity yield.
- **Spectators (N_{spec})**: Non-overlapping cold nuclear remnants. Continue at beam rapidity into Zero Degree Calorimeters (ZDCs).

Interactive Centrality Explorer

Open in your browser:

- [day2/day2_centrapidity_rapidity-explorer.html](#)

Physics of Centrality & Multiplicity Scaling

Two-Component Multiplicity Scaling

The total charged particle multiplicity at mid-rapidity follows the two-component Glauber model:

$$\frac{dN_{\text{ch}}}{d\eta} \approx n_{pp} \left[(1-x) \frac{N_{\text{part}}}{2} + x N_{\text{coll}} \right]$$

Where $x \approx 0.15$ at LHC energies.

- **Soft Processes** ($\propto N_{\text{part}}$): String fragmentation and low- Q^2 QCD processes.
- **Hard Processes** ($\propto N_{\text{coll}}$): Point-like perturbative QCD scatterings (jets, mini-jets), which dominate at high energies.

What Centrality Changes (and What It Does Not)

- **Amplitude $\uparrow$ (central)**: More participating nucleons $\Rightarrow$ higher $dN_{\text{ch}}/d\eta$. The curves stack vertically.
- **Shape (plateau width) unchanged**: The width σ_y of the rapidity distribution is set by the *beam energy* $\sqrt{s_{NN}}$, not by centrality. Normalized per participant pair, the distributions are self-similar.
- **Jacobian dip shallower in central**: Larger radial flow in central collisions boosts $\langle p_T \rangle$, increasing β and making the Jacobian suppression at $\eta = 0$ slightly less pronounced.

Landau Hydrodynamics & Rapidity Widths

Landau Model Rapidity Distribution (Sahoo pg. 145–146)

In Landau's hydrodynamic model of relativistic expansion, the rapidity distribution of produced pions is described by a Gaussian:

$$\frac{dN}{dy} = \frac{N}{\sqrt{2\pi\sigma_y^2}} \exp\left(-\frac{y^2}{2\sigma_y^2}\right)$$

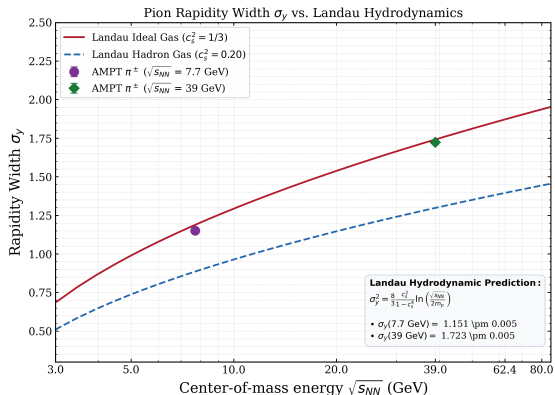
Where the width of the rapidity distribution σ_y relates to the center-of-mass energy and the speed of sound c_s in the medium:

$$\sigma_y^2 = \frac{8}{3} \frac{c_s^2}{1 - c_s^4} \ln\left(\frac{\sqrt{s_{NN}}}{2m_p}\right)$$

Landau Model Scaling Limits

- **Ideal Gas limit** ($c_s^2 = 1/3$): Prefactor becomes 1, so $\sigma_y = \sqrt{\ln(\sqrt{s_{NN}}/2m_p)}$.
- **Hadron Gas limit** ($c_s^2 \approx 0.20$): Premature freezing or lower speed of sound shrinks the rapidity width σ_y .

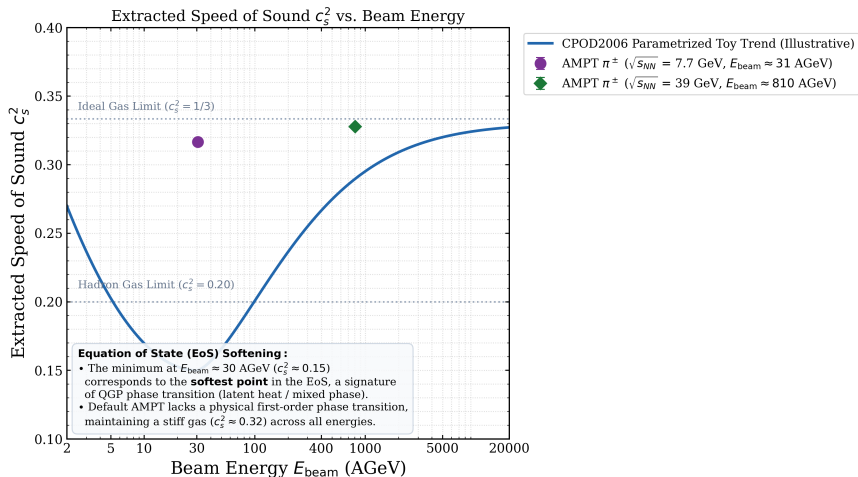
Pion Rapidity Widths: Landau Hydrodynamics



Width σ_y Analysis in AMPT

- The standard deviation σ_y of the pion rapidity distribution is calculated from AMPT at 7.7 and 39 GeV.
- Theoretical curves show Landau predictions for the Ideal Gas ($c_s^2 = 1/3$) and Hadron Gas ($c_s^2 = 0.20$).
- Observations:** AMPT pion rapidity widths are slightly wider and lie close to the Ideal Gas limit, indicating a stiff equation of state (EoS) in the default AMPT transport model.

Equation of State: Extracted Speed of Sound



Equation of State: Speed of Sound & EoS Softening

Speed of Sound $c_s^2 = \partial P / \partial \epsilon$ (Sahoo Fig. 5.16)

In Relativistic Hydrodynamics, c_s^2 characterizes the stiffness of the early medium. We extract c_s^2 from the pion rapidity width σ_y by inverting the Landau equation:

$$c_s^2 = \frac{-4 \ln(\sqrt{s_{NN}}/2m_p)}{3\sigma_y^2} + \sqrt{\left(\frac{4 \ln(\sqrt{s_{NN}}/2m_p)}{3\sigma_y^2}\right)^2 + 1}$$

Physical Interpretation & Motivation

- **The “Softest Point”:** Experimental data shows a clear minimum ($c_s^2 \approx 0.15$) at $E_{\text{beam}} \approx 30$ AGeV ($\sqrt{s_{NN}} \approx 7.7$ GeV).
- **Transition Signature:** This minimum is a classic signature of the transition from hadronic matter to QGP (mixed phase or onset of deconfinement).
- **AMPT Default:** Remains stiff ($c_s^2 \approx 0.32$), showing the lack of a physical first-order phase transition in default ZPC/ART.

Summary: Day 2 Hands-On Laboratory Exercises

Objective: Analytical & Data-Driven Rapidity Calculations

All exercises are framed in your Jupyter Notebooks located in the `exercises/` folder:

- `exercises/day02_hands_on.ipynb` (Student skeleton notebook)
- `exercises/day02_hands_on_solutions.ipynb` (Reference solutions with code and plots)

- 1 **Problem 1:** Calculate center-of-mass energy $\sqrt{s_{NN}}$ for RHIC Energy Scan values.
- 2 **Problem 2:** Prove why collider configurations have a massive kinematic advantage over fixed-target configurations.
- 3 **Problem 3:** Compute and verify Lorentz factors γ , velocities β , and beam rapidity limits y_{beam} for LHC Pb+Pb beams.
- 4 **Problem 4:** Parse AMPT events, extract particle species (pions, kaons, protons), and plot the mass-dependent mid-rapidity dip in $dN/d\eta$.
- 5 **Problem 5:** Extract the standard deviation (width σ_y) of pion rapidity and check compatibility with Landau's ideal gas limit vs. viscous gas models.